

Programme	B. Sc. Statistics				
Course Code	STA6CJ301(P)				
Course Title	Linear Regression Analysis				
Type of Course	Major				
Semester	VI				
Academic Level	300-399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites					
Course Summary Objective	Objective make students to describe and assess the strength of relationships between variables, to explain them using math model, check adequacy of model				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Student will be able to understand the context of regression analysis	U	C	Instructor-created exams / Quiz
CO2	Capable for fitting a linear regression model to the given data	Ap	P	Practical Assignment / Observation of Practical Skills
CO3	Able to scrutinize the fitted model using the model adequacy checking	Ap	P	Seminar Presentation / Group Tutorial Work
CO4	Capable for forecasting the future values using the fitted model	U	C	Instructor-created exams / Home Assignments
CO5	Able to understand which type of regression model (linear or non-linear) is suitable	Ap	P	One Minute Reflection Writing assignments
CO6		Ap	P	Viva Voce
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	Simple Regression		10	16
	1	Regression Model building: Scatter Diagram,		
	2	Regressor, Response, Error, uses of Regression.		
	3	Simple Linear Regression model,.		
	4	Assumptions, least square and maximum likelihood estimation of the parameters of the model.		
	5	Properties of least square estimators,		
	6	Hypothesis testing on slope and intercept of the model		
	7	Coefficient of Determination		
Sections from References:				
II	Multiple Regression		10	16
	8	Multiple Regression model, assumptions		
	9	least square and maximum likelihood estimation of the parameters of the model,		
	10	testing significance of regression coefficients, test on individual regression coefficient.		
	11	R^2 and adjusted R^2 .		
Sections from References:				
III			17	25
	12	Model adequacy checking		
	13	Residuals		
	14	Methods for scaling residuals,.		
	15	Residual plots,		
	16	PRESS statistic.		
	17	Detection and treatment of outliers		
Sections from References:				
IV			8	13
	18	Transformation and weighting to correct model inadequacy-variance stabilizing transformations		
	19	Transformations to linearize the model.		
	20	Concept of Box-Cox transformation..		
	21	Concept of multicollinearity,		
	22	Sources of multicollinearity, Variance Inflation Factor		
Sections from References:				
V	Open Ended Module: Practical Applications		30	
	1	Concept of non-linear regression, application to machine learning. Practical example of fitting a regression model using statistical software.		
Sections from References:				